Filippo Borsani

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EXPERIENCE

IDSIA USI-SUPSI — Mar. 2026 – Jul. 2026
Research Intern — Lugano, Switzerland

- Continued the thesis work toward a research publication by extending the autonomous surgery pipeline toward sim-to-real deployment across multiple robotic platforms and experimental setups.
- Built reproducible Isaac Lab training, benchmarking and evaluation pipelines for policy optimization, simulation experiments, quantitative analysis, and ablation studies.

IDSIA USI-SUPSI / ETH Zurich — Sep. 2025 – Feb. 2026
Master's Thesis Researcher — Lugano, Switzerland

- Developed Isaac Lab/Isaac Sim pipelines for autonomous ultrasound-guided spinal surgery using a Unitree G1 humanoid robot.
- Trained reinforcement learning policies with PyTorch for probe guidance and robotic vertebra drilling in physics-based simulation.
- Explored upper-body control strategies for safety-constrained humanoid fine manipulation.

PROJECTS

PX4 Quadcopter Control | *MATLAB/Simulink, ROS*

- Performed experimental system identification on a PX4 quadcopter, modeling component-level dynamics and reconstructing the full system from scratch.
- Designed model-based stabilization controllers and a Kalman filter for closed-loop flight control and state estimation in MATLAB/Simulink and ROS.

Autonomous Navigation Pipeline for Mobile Robots | *ROS, MATLAB/Simulink, Gazebo*

- Implemented a complete autonomy stack for a TurtleBot3, including SLAM, global path planning (A*, RRT*), obstacle-aware navigation, and closed-loop trajectory tracking.
- Integrated and validated the full pipeline in Gazebo simulation before deployment on a physical TurtleBot using ROS and MATLAB/Simulink.

EDUCATION

Politecnico di Milano — Sep. 2023 – Mar. 2026
M.Sc. in Mechanical Engineering, Mechatronics & Robotics — Milano, Italy

- Grade: 110/110; focus on robotics, control systems, system identification, optimization, and mechatronic design.

TU Delft — Feb. 2025 – Jul. 2025
Erasmus Program, Robotics, Systems and Control — Delft, Netherlands

- Coursework and projects in machine learning, advanced control, autonomous systems, and robotic modeling.

Politecnico di Milano — Sep. 2020 – Sep. 2023
B.Sc. in Mechanical Engineering — Milano, Italy

- Grade: 106/110; Best Freshmen Prize A.Y. 2020/21.

SKILLS

Programming : Python, PyTorch, JAX, MATLAB/Simulink, Git, Linux/Ubuntu
Tools : ROS, Isaac Lab, Isaac Sim, COMSOL, Inventor, SolidWorks, CATIA, CAD/FEM
Languages : Italian native, English proficient